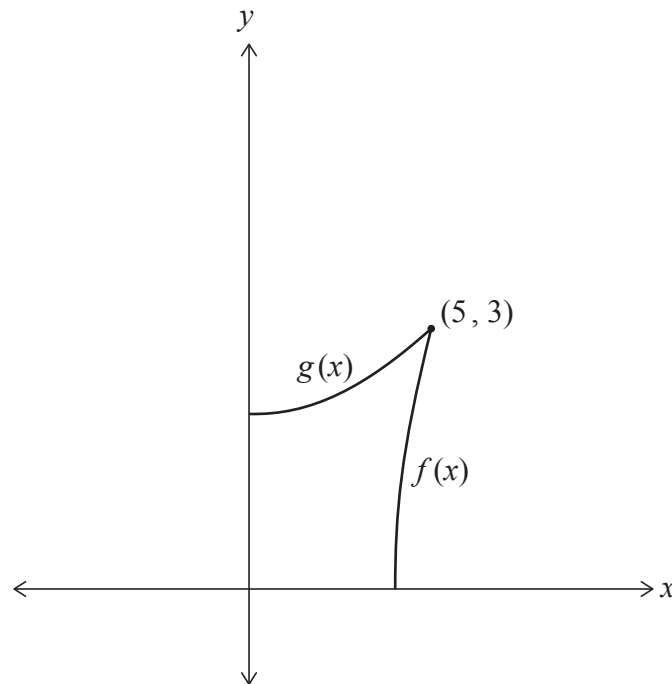


8. [Maximum mark: 6]

The following diagram shows the graphs of two functions,  $f$  and  $g$ , where

$$f(x) = \sqrt{x^2 - 16} \text{ for } 4 \leq x \leq 5, \text{ and}$$

$$g(x) = 2 + \frac{x^2}{25} \text{ for } 0 \leq x \leq 5.$$



The graphs of  $f$  and  $g$  intersect at the point  $(5, 3)$ .

The region bounded by the graphs of  $f$ ,  $g$  and both axes is rotated  $360^\circ$  about the  $y$ -axis to form a solid of revolution.

Find the volume of the solid formed.

**(This question continues on the following page)**



(Question 8 continued)

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Turn over